

Reply to A.W. Chan et al

Authors: [Erin F. Gillespie, MD, MPH](#) , [Max Vayntrub, MD](#) , and [Jonathan T. Yang, MD, PhD](#) | [AUTHORS INFO & AFFILIATIONS](#)

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ACCOMPANYING ARTICLE

 CORRESPONDENCE

[Prophylactic Radiation Therapy for High-Risk Asymptomatic Bone Metastases: A New Standard of Care or Need for More Data?](#)

We appreciate Chan et al¹ for highlighting several important points as we consider the findings from our recently published phase II randomized trial² of radiation therapy (RT) to reduce skeletal-related events (SREs) among patients with high-risk asymptomatic bone metastases. We agree that a confirmatory phase III clinical trial is needed to warrant practice change and herein provide insights into a future study design that is under development.

Understanding the patient population is critical for interpreting trial findings, and we can provide additional context in this regard. First, the eligibility criteria specified >5 metastatic sites to avoid overlap with ongoing investigations in oligometastatic and oligoprogressive disease. Specifically, we were simultaneously conducting a randomized trial of radiation for oligoprogressive disease that was recently presented by Tsai et al³ that did not show an overall survival (OS) benefit with radiation. Second, we excluded patients with unstable disease requiring surgical intervention by incorporating elements of Mirel's criteria⁴ for the peripheral skeleton in the eligibility criteria (ie, >2/3 cortical thickness); in addition, it was not possible to obtain a Spinal Instability Neoplastic Score score in the unstable range (13 or greater)⁵ without symptoms and vertebral body collapse/fracture which were also exclusion criteria. When considering the provocative finding of prolonged OS, it is imperative to remember that this was an unpowered secondary analysis. Although stratification variables can further prevent imbalance introduced by chance in a randomized trial, the sample size and event rate limit the number of variables that are feasible from a statistical perspective. For a primary end point of SREs in the current trial as well as a proposed future phase III concept, the use of bone-modifying agents has been the factor of greatest concern (49% in the RT arm v 51% in the standard-of-care [SOC] arm, as previously published) followed by the number of high-risk bone metastases enrolled which favored the SOC arm (41% in the RT arm v 23% in the SOC arm, as previously published).

Selecting and defining the primary outcome is essential to determining the initial success or failure of a clinical trial. SRE by definition includes pathologic fractures, spinal cord compression, and intervention with surgery or radiation. For our phase II trial, SRE was selected because of the impact of such events on patient-centered outcomes, particularly at the end of life, and was supported by previous randomized trials of bone-modifying

agents that validated its use.⁶ In anticipation of criticism regarding the importance of early radiation to prevent future radiation, we did conduct an analysis omitting radiation for pain—this was still highly significant (12% v 0%, $P = .008$). We do, nonetheless, have concerns that pain alone can affect functional status and morbidity so would caution against omitting this outcome entirely.

The heterogeneity of multiple radiation regimens has also been a common point for consideration. This approach was selected for two main reasons: (1) the absence of existing evidence guiding appropriate selection of radiation dose in the *asymptomatic* setting (compared with the symptomatic setting where the primary goal is reduction of pain) and (2) to facilitate enrollment via a pragmatic approach. Although we agree that stereotactic RT can potentially lead to higher risk of fracture, the end point of SREs would have captured these events. Furthermore, 27 Gy in three fractions was used per institutional standard rather than a single-fraction regimen.⁷ While one proposal for a future phase III trial is to limit radiation to 8 Gy $\times$ 1, we continue to advocate for a pragmatic approach given the low rate of SREs and adverse events in the treatment arm regardless of regimen. Ultimately, the current trial and future concepts are probing whether *any* radiation is effective in this novel setting, when applied with the radiation regimen deemed safe and appropriate given the heterogeneity present among bone metastases.

A key aspect for future implementation research will be identification of patients at high risk of SREs. We did not, however, standardize *screening* imaging and instead leveraged existing images already indicated in routine care, including either computed tomography or positron emission tomography. This is a fundamentally different study design than the PROMPTS trial, which found that screening spine magnetic resonance imaging among patients with castrate-resistant prostate cancer nonsignificantly reduced clinical spinal cord compression from 6.7% versus 4.3% (difference, -2.4% [95% CI, -4.2 to 0.1]) at 1 year.⁸

We appreciate Chan et al,¹ for encouraging ongoing discussion about tradeoffs that will inform the optimal design of future phase III trials, which will be needed to for this work to justify a new SOC.

Publisher's Note

The article by Gillespie et al entitled "Reply to A.W. Chan et al" (*J Clin Oncol* 10.1200/JCO.23.02566) was published online February 6, 2024, with an error.

Co-author **Max Vaynrub, MD** was listed twice. He should have been included as the second author only.

This has been corrected as of February 26, 2024. *Journal of Clinical Oncology* apologizes for the error.

Authors' Disclosures of Potential Conflicts of Interest

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Erin F. Gillespie

Other Relationship: eContour

Max Vaynrub

Honoraria: Leiden University Medical Center (Inst)

Jonathan T. Yang

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